

Time Series Forecasting & Portfolio Optimization using NSE Stocks

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1. Executive Summary

This project focuses on applying Time Series Analysis techniques for forecasting stock prices of selected NSE-listed companies and constructing optimized portfolio allocation strategies using quantitative methods. The analysis was performed on five major Indian companies from diversified sectors: Banking, Information Technology, Pharmaceuticals, FMCG, and Automobile.

Historical stock data from January 2021 to May 2026 was collected using [Yahoo Finance](#) and analyzed using statistical and forecasting techniques. Three forecasting models – ARIMA, Prophet, and Holt-Winters Exponential Smoothing – were implemented to predict future stock prices and evaluate forecasting performance using RMSE, MAPE, and Directional Accuracy.

The project also included volatility analysis, stationarity testing, seasonal decomposition, return distribution analysis, and correlation analysis. Based on the forecasting outputs and risk analysis, multiple portfolio allocation strategies were designed using forecast-based weighting, volatility minimization, and diversification principles.

The results demonstrated that different models performed differently across sectors. Prophet produced superior performance for TCS, while Holt-Winters achieved lower forecasting error for HDFC Bank and Eicher Motors. Portfolio strategies based on forecasted returns indicated higher allocation toward HDFC Bank and ITC due to strong positive predicted returns.

This capstone project successfully combines financial analytics, machine learning concepts, statistical forecasting, and portfolio optimization techniques to simulate real-world quantitative investment decision-making.

2. Introduction

Financial markets generate massive amounts of time-dependent data, making Time Series Analysis one of the most important analytical techniques in quantitative finance. Stock prices continuously fluctuate due to macroeconomic conditions, company performance, investor

sentiment, sectoral trends, and market volatility. Forecasting stock prices accurately can help investors improve portfolio performance and manage investment risk effectively.

Time Series Analysis involves studying historical observations collected over time to identify underlying patterns such as trends, seasonality, cyclicalities, and volatility. In financial markets, forecasting models are widely used for predicting stock prices, measuring risk, estimating future returns, and supporting portfolio construction decisions.

This project aims to apply advanced forecasting models to historical stock market data from the National Stock Exchange (NSE) of India and evaluate the performance of multiple forecasting approaches. The study also integrates portfolio optimization strategies to allocate investment capital efficiently based on expected returns, volatility, and diversification principles.

The project combines concepts from:

- **Quantitative Finance**
- **Statistical Forecasting**
- **Risk Management**
- **Portfolio Theory**
- **Financial Data Analytics**
- **Machine Learning for Time Series**

3. Objectives of the Project

The major objectives of this project are:

- **To analyze historical NSE stock price data using Time Series techniques**
- **To identify trends, volatility, seasonality, and correlation patterns**
- **To test stationarity and preprocess financial time series data**
- **To build forecasting models using ARIMA, Prophet, and Holt-Winters methods**
- **To compare forecasting performance using multiple evaluation metrics**
- **To predict short-term future stock prices**
- **To design portfolio allocation strategies using quantitative methods**
- **To evaluate risk-return tradeoffs in portfolio construction**
- **To simulate practical investment decision-making using data-driven analysis**

4. Dataset Description

The dataset used in this project was obtained from [Yahoo Finance](#) using the Python library `yfinance`.

Selected Stocks

The following NSE-listed companies were selected from different sectors to ensure diversification:

Stock	Sector
HDFC Bank	Banking
Tata Consultancy Services	Information Technology
Sun Pharmaceutical Industries	Pharmaceuticals
ITC Limited	FMCG
Eicher Motors	Automobile

Dataset Specifications

- Data Frequency: Daily
- Time Period: January 1, 2021 to May 10, 2026
- Variables Used:
 - Open
 - High
 - Low
 - Close
 - Volume

5. Data Preprocessing

Financial time series data often contains noise, trends, and non-stationary behavior. Therefore, several preprocessing techniques were applied before model implementation.

Data Cleaning

The following preprocessing steps were performed:

- Removal of unnecessary columns
- Handling missing values
- Date index formatting
- Flattening multi-index columns
- Conversion into structured time series format

Return Calculation

Daily percentage returns were calculated for each stock using closing prices.

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100$$

Where:

- R_{tR_tRt} = Daily Return
- P_{tP_tPt} = Current Closing Price
- $P_{t-1P_t-1Pt-1}$ = Previous Closing Price

Data Splitting

The dataset was divided into training and validation periods:

Dataset	Period
Training Data	Jan 2021 – Dec 2025
Validation Data	Jan 2026 – May 2026

Each stock contained:

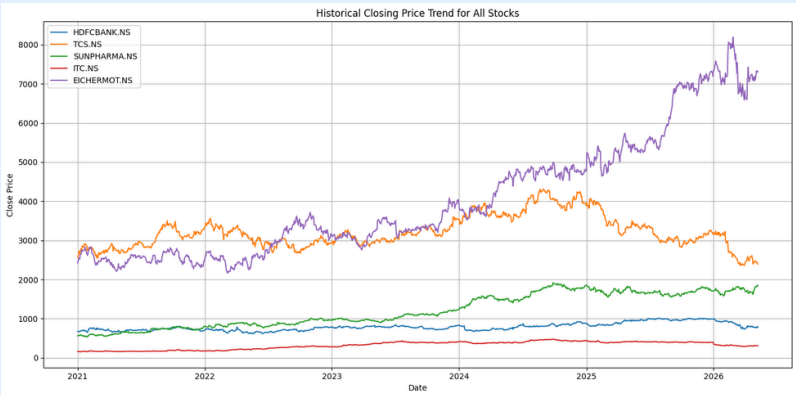
- Training observations: 1236
- Validation observations: 86

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand stock behavior, volatility patterns, and relationships among different assets.

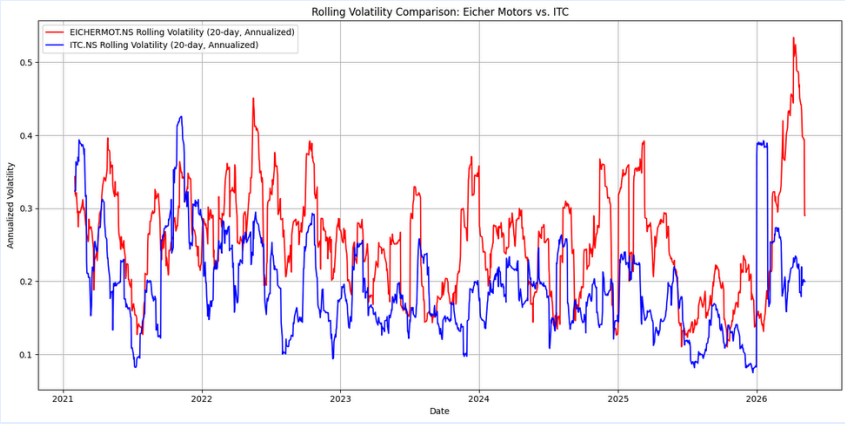
Historical Price Trend Analysis

Line plots of stock closing prices revealed long-term growth patterns and sectoral differences. Stocks such as Eicher Motors and TCS demonstrated strong upward price trends, while ITC showed comparatively stable movement.



Rolling Volatility Analysis

A 20-day rolling standard deviation was used to measure short-term volatility fluctuations.



Key observations:

- Eicher Motors exhibited the highest volatility.
- ITC showed relatively stable volatility behavior.
- Banking and automobile sectors experienced higher fluctuations during uncertain market periods.

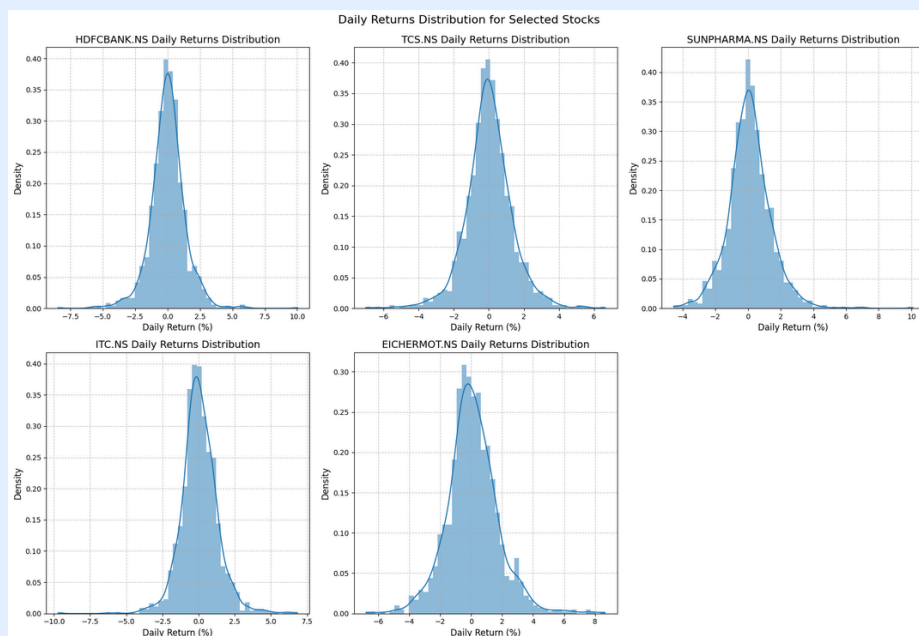
Return Distribution Analysis

Histograms and KDE plots were used to analyze daily return distributions.

Descriptive Statistics of Returns

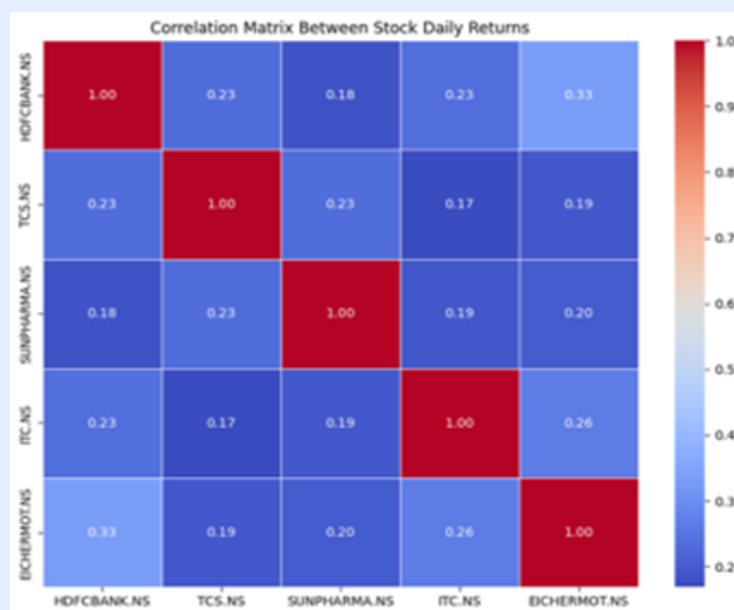
Stock	Mean Return	Std Dev	Skewness	Kurtosis
HDFCBANK.NS	0.0204	1.3652	0.0611	5.1739
TCS.NS	0.0037	1.3446	-0.0642	2.7402
SUNPHARMA.NS	0.0989	1.3355	0.5922	4.0396
ITC.NS	0.0559	1.2938	0.0006	5.0633
EICHERMOT.NS	0.0975	1.6770	0.4199	2.3510

The positive kurtosis values indicate fat-tailed distributions, suggesting the presence of extreme market movements.



Correlation Analysis

Correlation analysis was performed to understand relationships among stock returns.



Key observations:

- Moderate correlations existed among stocks.
- Pharma and IT sectors showed relatively lower correlation.
- Diversification benefits were possible due to sectoral differences.

7. Stationarity Testing

Stationarity is a critical assumption in many time series forecasting models.

The Augmented Dickey-Fuller (ADF) Test was applied to determine whether stock price series were stationary.

ADF Test on Original Prices

All stocks produced p-values greater than 0.05, indicating non-stationary behavior.

Example:

- HDFCBANK.NS: p-value = 0.585
- TCS.NS: p-value = 0.181

This confirmed that raw stock prices contained trends and changing statistical properties over time.

First Differencing

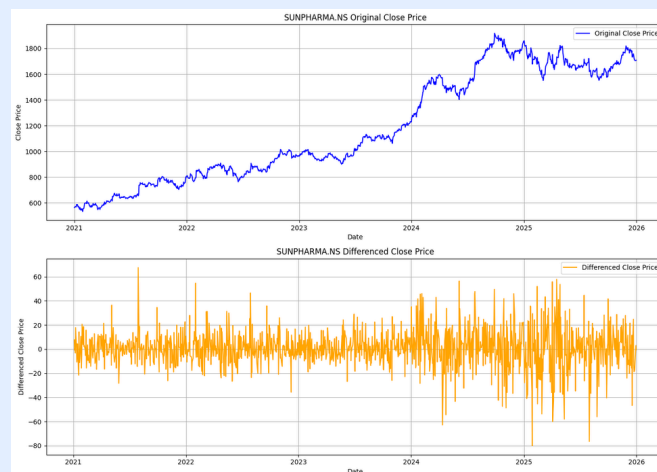
To achieve stationarity, first-order differencing was applied.

$$Y'_t = Y_t - Y_{t-1}$$

After differencing, all stocks produced p-values below 0.05, confirming stationarity.

Example:

- HDFCBANK.NS: 3.298×10^{-26}
- TCS.NS: 0.0000



8. Forecasting Models

Three forecasting models were implemented and compared.

8.1 ARIMA Model

The ARIMA model combines autoregressive, differencing, and moving average components.

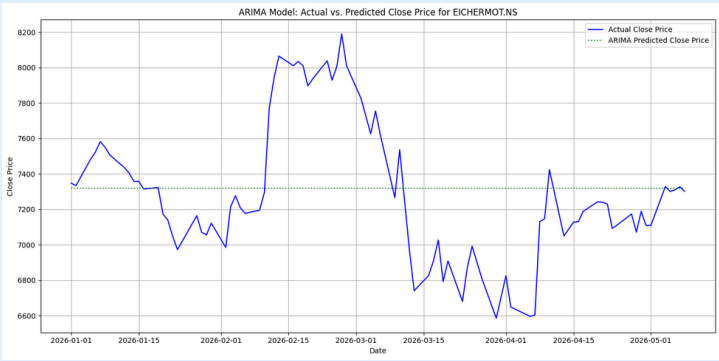
ARIMA(p,d,q)ARIMA(p,d,q)ARIMA(p,d,q)

The model order selected was:

- $p = 1$
- $d = 1$
- $q = 1$

ARIMA Performance

Stock	RMSE	MAPE	Directional Accuracy
HDFCBANK.NS	147.18	15.68%	9.41%
TCS.NS	524.52	17.48%	40.00%
SUNPHARMA.NS	65.68	3.00%	5.88%
ITC.NS	82.44	26.21%	12.94%
EICHERMOT.NS	383.48	4.11%	12.94%



The ARIMA model performed reasonably well for short-term forecasting but struggled with directional prediction accuracy.

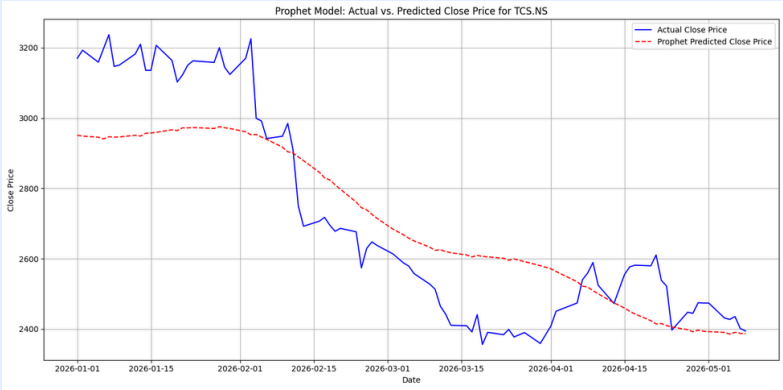
8.2 Prophet Model

The Prophet model is designed for handling trend and seasonality efficiently.

It showed superior forecasting performance for certain stocks such as TCS.

Prophet Performance

Stock	RMSE	MAPE	Directional Accuracy
HDFCBANK.NS	187.46	20.17%	56.47%
TCS.NS	154.64	4.91%	58.82%
SUNPHARMA.NS	110.60	5.13%	48.24%
ITC.NS	54.02	17.15%	54.12%
EICHERMOT.NS	453.07	5.34%	47.06%



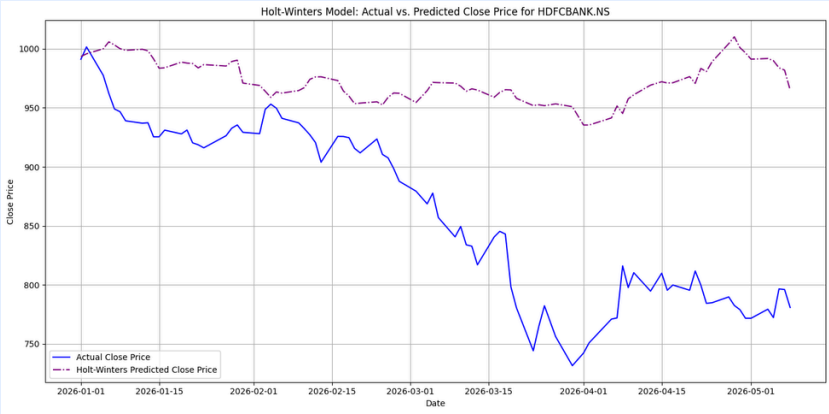
The Prophet model achieved the highest directional accuracy among all models.

8.3 Holt-Winters Exponential Smoothing

The Holt-Winters model captures trend and seasonality using smoothing parameters.

Holt-Winters Performance

Stock	RMSE	MAPE	Directional Accuracy
HDFCBANK.NS	129.83	13.51%	54.12%
TCS.NS	504.42	17.00%	52.94%
SUNPHARMA.NS	100.25	4.96%	52.94%
ITC.NS	73.95	23.46%	57.65%
EICHERMOT.NS	373.40	4.17%	57.65%



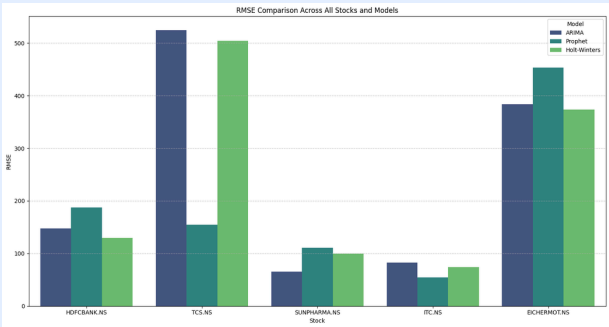
Holt-Winters achieved lower RMSE values for HDFC Bank and Eicher Motors.

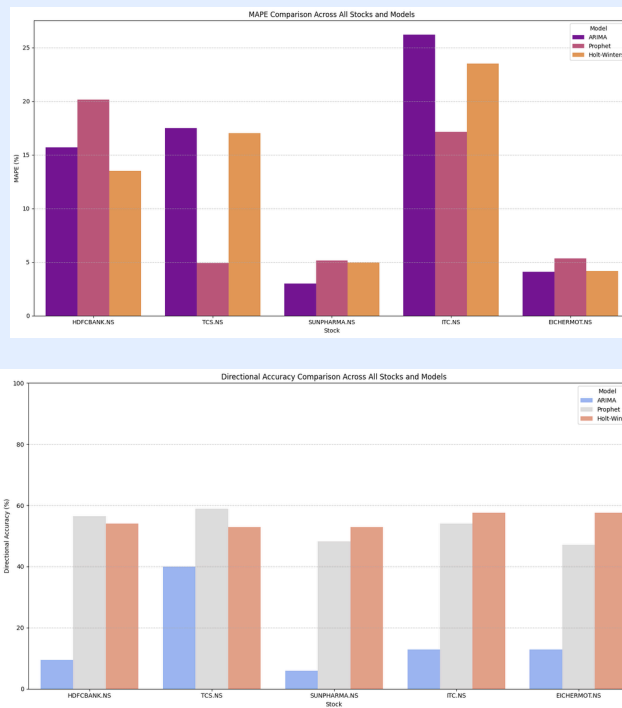
9. Model Comparison

The forecasting models were compared using RMSE, MAPE, and Directional Accuracy.

Key findings:

- Prophet achieved superior directional forecasting capability.
- Holt-Winters produced lower prediction errors for certain stocks.
- ARIMA required stationarity transformation but showed stable forecasting behavior.





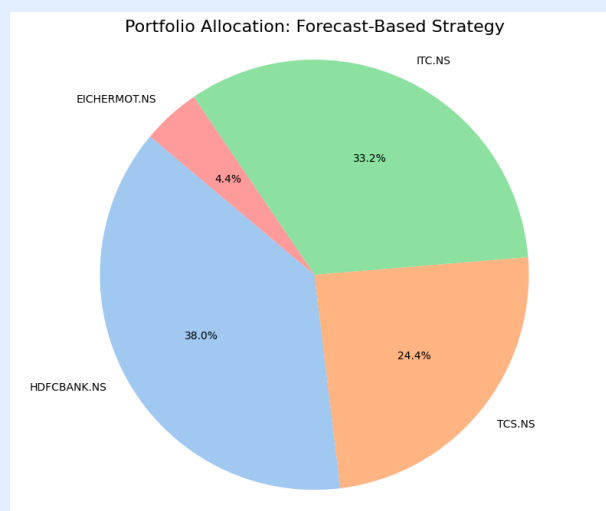
Overall:

- Best RMSE for HDFC Bank → Holt-Winters
- Best RMSE for TCS → Prophet
- Best Directional Accuracy → Prophet & Holt-Winters

10. Portfolio Construction Strategies

A virtual investment capital of ₹10,00,000 was allocated using quantitative portfolio construction techniques.

Average forecast prices from all three models were used to estimate expected returns.



10.1 Forecast-Based Allocation Strategy

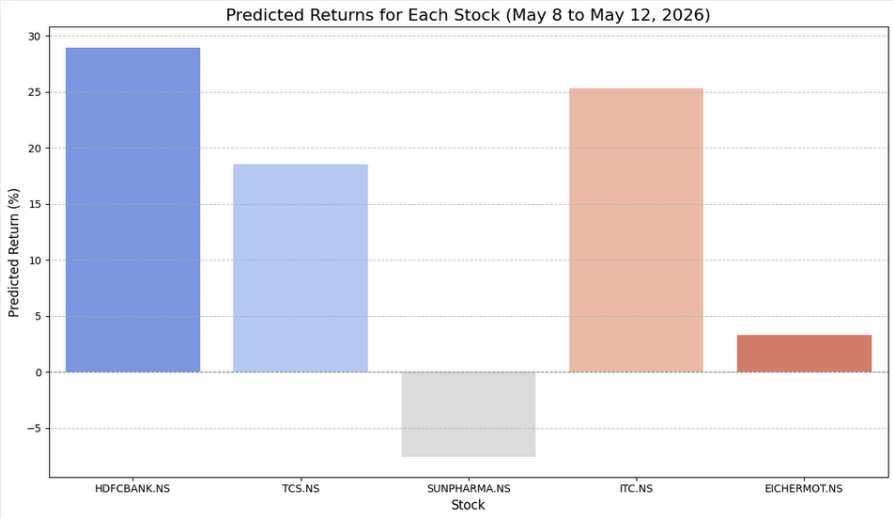
Stocks with higher predicted returns received larger allocations.

Predicted Returns

Stock	Predicted Return
HDFCBANK.NS	28.97%
TCS.NS	18.56%
SUNPHARMA.NS	-7.52%
ITC.NS	25.29%
EICHERMOT.NS	3.33%

Allocation Results

Stock	Allocation
HDFCBANK.NS	₹380,445.89
TCS.NS	₹243,714.49
SUNPHARMA.NS	₹0
ITC.NS	₹332,088.12
EICHERMOT.NS	₹43,751.51

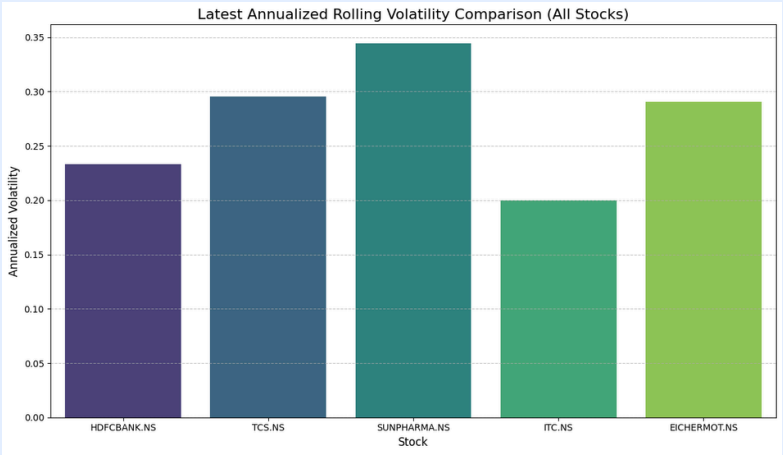


HDFC Bank and ITC received the highest weights due to strong expected returns.

10.2 Volatility-Based Allocation

This strategy allocated capital inversely proportional to annualized volatility.

Lower volatility stocks received higher allocations.

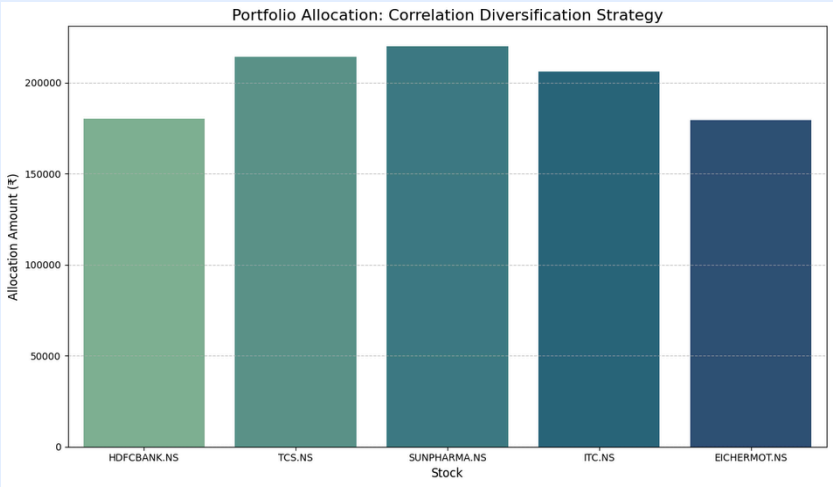


ITC received the highest allocation due to its relatively stable behavior.

10.3 Correlation Diversification Strategy

Capital allocation was performed inversely proportional to average correlation.

Stocks with lower correlation received higher portfolio weights to maximize diversification benefits.



SUNPHARMA and TCS received higher allocations due to lower average correlations.

11. Key Findings and Insights

The major findings from the project are:

- Stock prices were non-stationary and required differencing.
- Prophet achieved superior directional forecasting accuracy.
- Holt-Winters performed effectively for stable trend forecasting.
- Eicher Motors exhibited the highest volatility among selected stocks.
- ITC showed stable return behavior with lower volatility.
- Diversified sectoral allocation reduced portfolio concentration risk.
- Correlation-based allocation improved diversification efficiency.

12. Challenges Faced

Several practical challenges were encountered:

- Financial markets are highly unpredictable.
- Sudden market movements reduced forecasting accuracy.
- Non-stationary stock prices required transformation.
- Model performance varied significantly across sectors.
- Short-term forecasting remained difficult during volatile periods.

13. Conclusion

This project successfully demonstrated the application of Time Series Analysis techniques for financial forecasting and portfolio optimization using NSE stock data. Advanced statistical and forecasting models such as ARIMA, Prophet, and Holt-Winters were implemented to analyze trends, volatility, and future stock movements.

The study revealed that no single forecasting model consistently outperformed others across all stocks, highlighting the importance of model comparison and ensemble thinking in financial analytics. Portfolio allocation strategies based on expected returns, volatility, and correlation further demonstrated the significance of balancing profitability with risk management.

Overall, the project provided valuable insights into quantitative finance, forecasting methodologies, and real-world investment decision-making. It also strengthened practical skills in Python programming, financial analytics, statistical modeling, and portfolio management.

14. References

- [Google Drive](#)